ENN581
ROBOT MOTION, CONTROL AND PLANNING



Week 1
Introduction to mechanisms

ENN581 Robot Motion, Control and Planning

Dr. Krishna Manaswi Digumarti

School of Electrical Engineering and Robotics



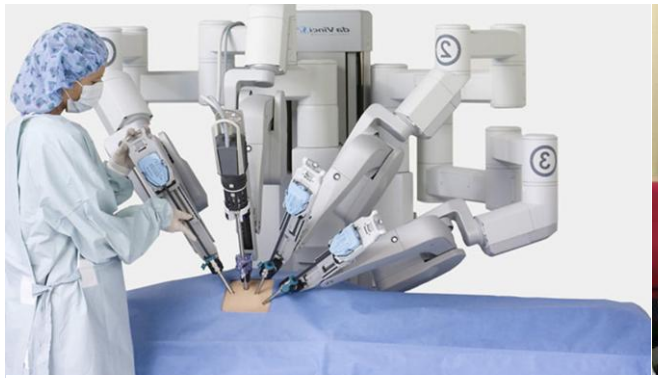
ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.



Robots are everywhere



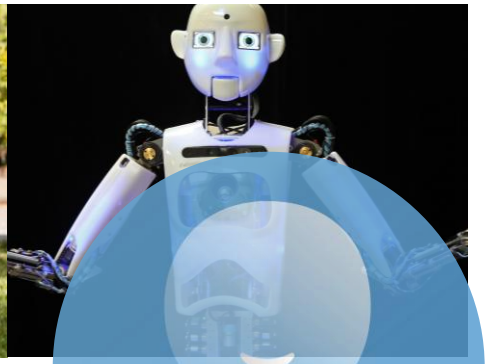




Don’t you want to orchestrate their motions?





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This unit will teach you the building blocks of robot motion

Learning outcomes:

1. Critically analyse and reflect on the kinematics of mobile and manipulator arm robots
2. Critically analyse and reflect on the dynamics of mobile and manipulator arm robots
3. Design and implement programming and simulation knowledge to complex robotics problems

Content:

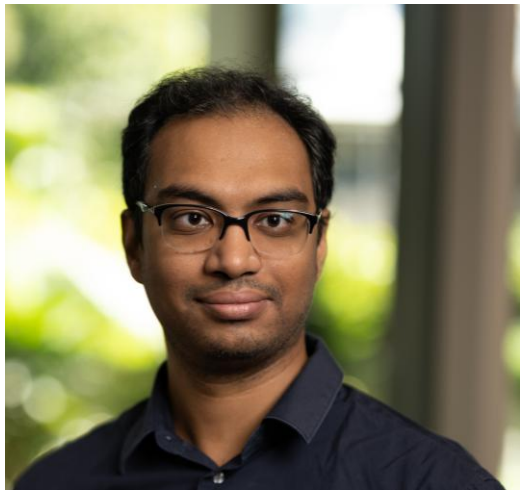
Kinematics, dynamics, methods of control, path planning

Refer to the unit's page on HiQ: link here



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Teaching team





Dr. Krishna Manaswi Digumarti
Unit coordinator and lecturer
krishnamanaswi.digumarti@qut.edu.au

Dr. Payam Nori Zadeh
Teaching Assistant

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You'll meet Dr. Tobias Fischer in Weeks 1 and 2



In the first two weeks, Dr. Fischer **will not deliver a regular lecture**.
Please listen to my recording and discuss the material with him in the class.
Bring questions about the material, latest techniques and anything relevant to the week's content.
Standard classes resume from week 3.
Practicals will run as usual in all weeks with Dr. Payam Nori Zadeh.

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Lectures and practical sessions

Activity	Weeks	Time	Location
Lectures	1-13	Monday, 13:00 to 15:00	GP-Z-606
Practical	1-13	Monday, 16:00 to 18:00	GP-S-620



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Typical weekly plan

Day	What you can expect from me	What is expected of you
Friday (prev. week)	Canvas announcement on the plan for the week – topics, practical, assessments, etc. Lecture slides uploaded to canvas.	Check your email and canvas for this announcement. Peruse suggested material. Prepare for practical.
Monday	Lecture in **GP-Z-606 13:00-15:00**, Practical **GP-S-620 16:00 to 18:00**. Release of assessments and solutions, if any.	Attend the lecture and practical. Watch previous week's lecture if you missed it, before class.
Tuesday	Updated lecture notes uploaded to Canvas. Recorded lecture is available on Canvas.	Watch the recorded lecture and attempt homework questions. Work on assessments.
Any day	Important changes will be communicated through announcements on Canvas.	Check email and Canvas for announcements.



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Canvas resources

Access Canvas at https://canvas.qut.edu.au/

- Announcements - check the unit page and your emails.
- Documents - slides, notes, lectures and videos.
- Reading material - books, papers and library resources.
- Assessment details - instructions and submissions.
- Unit outline and schedule of topics.



Assessments

Assessment	Takes place in	Weight [%]
Mid semester exam	Week 7	20
Problem solving tasks	Throughout the semester	40
Examination	Exam period	40

All assessments are evaluations of individual work.
Pay attention to the deadlines for individual assessments on Canvas.



Use of Generative Artificial Intelligence (AI) tools

The use of generative artificial intelligence (AI) tools is allowed to help you study and find out more about the topic; noting AI-generated content may not be accurate, reliable, or may display bias. Any content generated may not be copied and included in your final submission. This is because the task has been designed for you to demonstrate your achievement of the learning outcomes of the unit. Go to Learning with generative AI for additional information.

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Requests for extension

- The automatically approved 48h extension is applicable ONLY to problem solving tasks (programming tasks)
- The 48h extension option is NOT available for: Mid-Semester Exam and the Final Exam
  - For those, you need to submit a request for an extension based upon special circumstances along with supporting documentation.
- For all assessments: extension due to special circumstances requires supporting documents
  - Students must submit their request before the due date whenever possible, or in some cases immediately after the due date/time if they can prove that exceptional circumstances prevented them from applying beforehand.



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Exam Prep Tips: Week 1

Consistency is key

- Develop habits to help you be consistent with your study. This might include:
  - Time management strategies
  - Getting quality sleep
  - Eliminating study distractions
- Watch **this video** (6 min.) to learn how habit formation works.



Start a 'cheat' sheet

- One page that summarises the main ideas and formulas for the exam.
- Helpful, even if not permitted in the exam.
- Watch **this video** (9 min.) to learn how to make a great cheat sheet.



See the full list of tips in the **Guide to Succeeding in Engineering Exams**

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Recommended books

- Robotics, Vision and Control - Fundamental Algorithms in Python by Peter Corke (link)
- Introduction to Robotics: Mechanics and Control by John J. Craig - (link)
- Modern robotics: Mechanics, Planning and Control by Kevin M. Lynch and Frank C. Park (link)
- Introduction to Autonomous Mobile Robots by Roland Siegwart, Illah R. Nourbakhsh and Davide Scaramuzza (link)
- Planning Algorithms by Steven M LaValle (link)

- Other reading material will be suggested in the class and on canvas.



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Communication

- Check the resources on Canvas.
- Check the discussions section on Canvas.
- If your questions has not been answered, create a new post on Canvas.
  Do not email the instructor.
- If you can answer your colleague's questions, please feel free to do so.
  Note: the teaching staff does not take any responsibility for a student's answer, unless it was clearly and explicitly confirmed as being accurate by a member of this unit's teaching staff.
- Consider using a direct private email if your issue is personal and individual. Make sure that the emails clearly indicate the unit code ENN581 in the subject line for a prompt response. Including your full name and student number can help in certain cases.



ENN581 in the context of EN52 / 72

EN52 Master of Robotics and Artificial Intelligence 2023

Year 1				Year 2
Semester 1		Semester 2		Semester 1
48 Credit Points		48 Credit Points		48 Credit Points
CAB420 Machine Learning 12CP	→	**ENN583** Foundations of Robotics Vision 12CP	→	**ENN585** Advanced Machine Learning 12CP
EGH437 Robot Anatomy 12CP	→	**ENN581** Robot Motion, Control and Planning 12CP	→	**ENN519** Entrepreneurship and Applications 12CP
EGH431 Principles of Control Engineering 6CP	→	**ENN582** Reinforcement Learning and Optimal Control 6CP	→	**ENN584** Robot Systems 12CP
EGH432 Foundations of Kinematics and Algorithms in Robotics 6CP		**ENN586** Decision and Control 6CP		
ENN541 Research Methods for Engineers 12CP	→	**ENN595-1** Project 1 12CP	→	**ENN595-2** Project 2 12CP

→ Pre-Requisites

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Introduction to Mechanisms and Concepts of Kinematics

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Kinematics

- Kinematics is the study of motion without considering the forces that cause it
- We are interested only in the geometry of motion – how things move



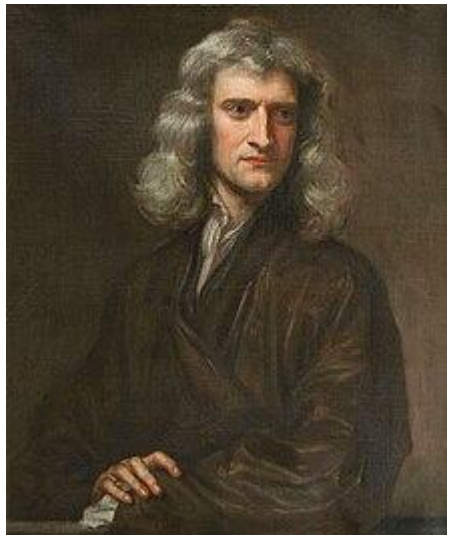
André-Marie Ampère

Cinématique - Essai sur la Philosophie des Sciences, 1834

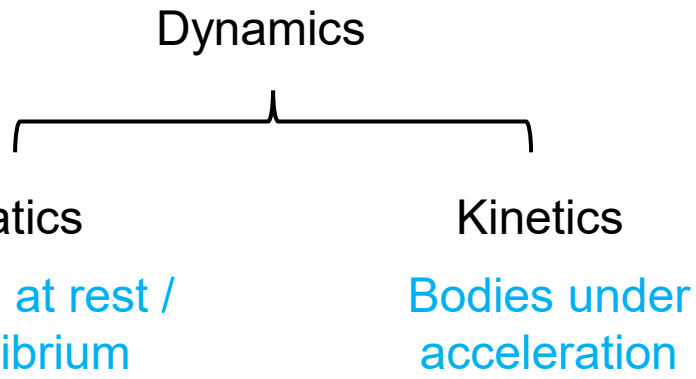
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Dynamics

- Dynamics is the study of the forces required to cause motion
- Why are objects moving or not moving?



Sir Isaac Newton





Gottfried Wilhelm von Leibniz

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Today's lecture

- What are mechanisms?
- What does degrees of freedom mean?
- What are various types of joints?
- How many actuators do we need in a robot arm?
- Where all can a robot arm reach?

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Mechanism



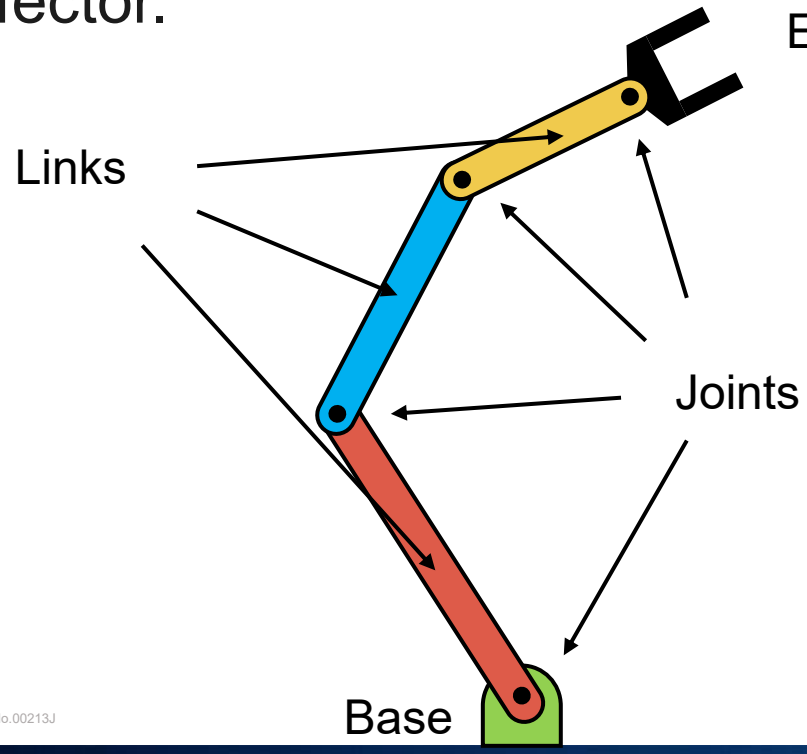
“A mechanism is a mechanical system that transfers/modifies motion and/or forces from one or more sources to one or more outputs.”



Kinematic Chain

A kinematic chain is a system of rigid bodies connected by joints.

A mechanism is a collection of one or more kinematic chains connecting the base to the end effector.



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A typical robot manipulator arm is a mechanism



Link

Joint – a connection between links

End-effector or tool. Interacts with the world

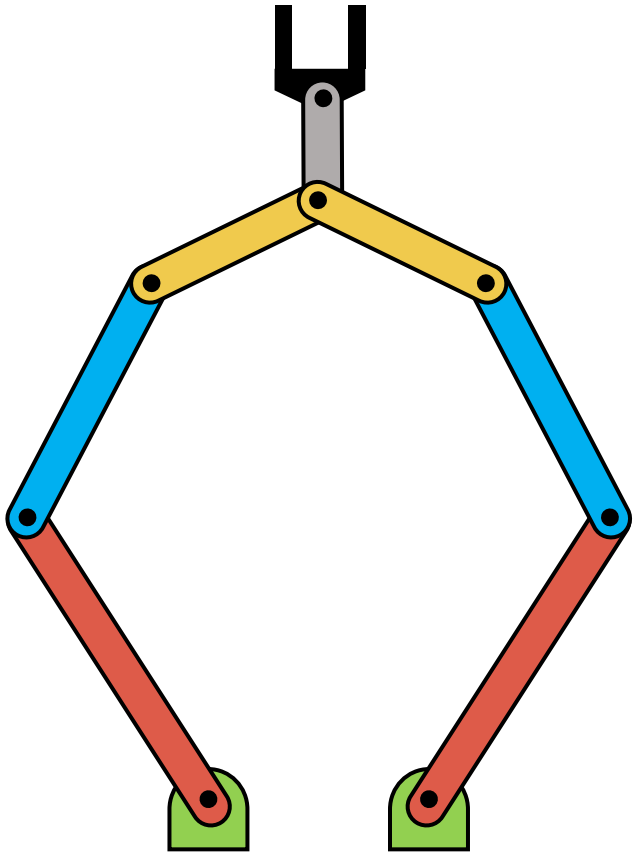
Base – rigidly connected to the ground

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Serial vs parallel chains



Serial



Parallel



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Open vs Closed chain

- A chain is called **closed** if it forms a closed loop.
- A chain that is not closed is called an **open** chain.
- If each link of an open chain except the first and the last link is connected to two other links it is called a serial chain.

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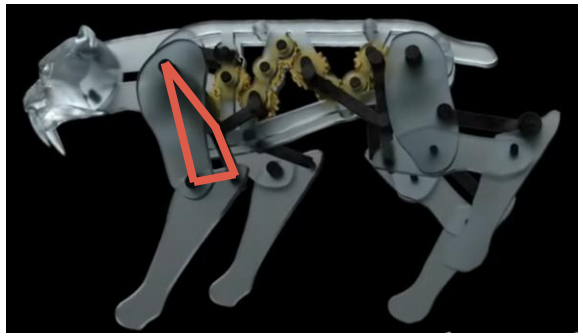
Closed loop mechanisms

Advantages of closed loop mechanisms

- Actuators at the base make the robot lighter
- Larger load can be carried – mechanical advantage
- Interesting motion paths can be obtained



Strandbeesten by Theo Jansen



Cyber Tiger by S. Coros et al. 2013

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Degrees of freedom

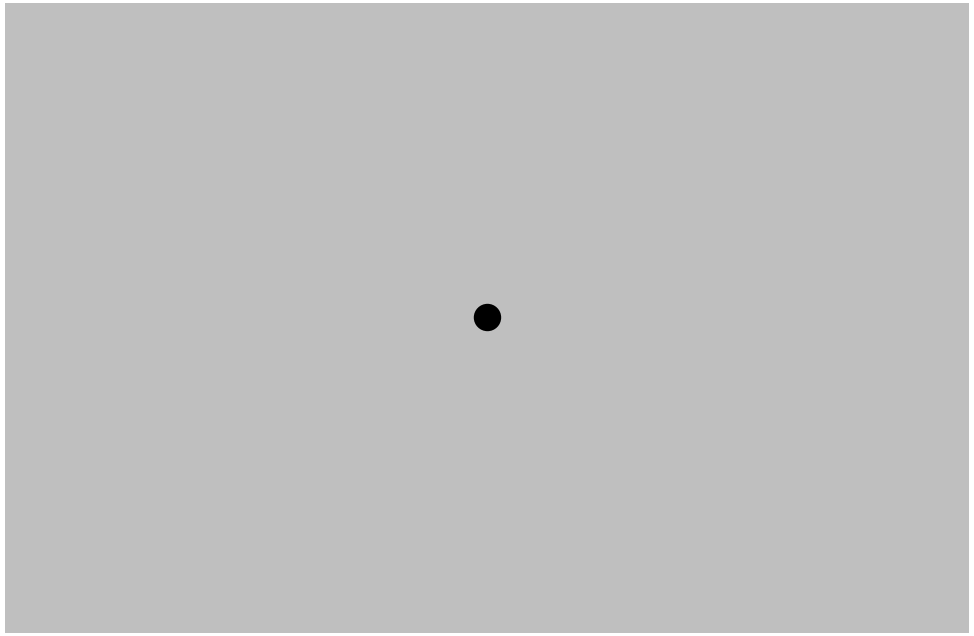
The number of independent parameters that define the configuration of a mechanism is called its degrees of freedom (DOF)

How many quantities do you need to completely define the system, at a given moment in time?

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Example: A point in 2D space

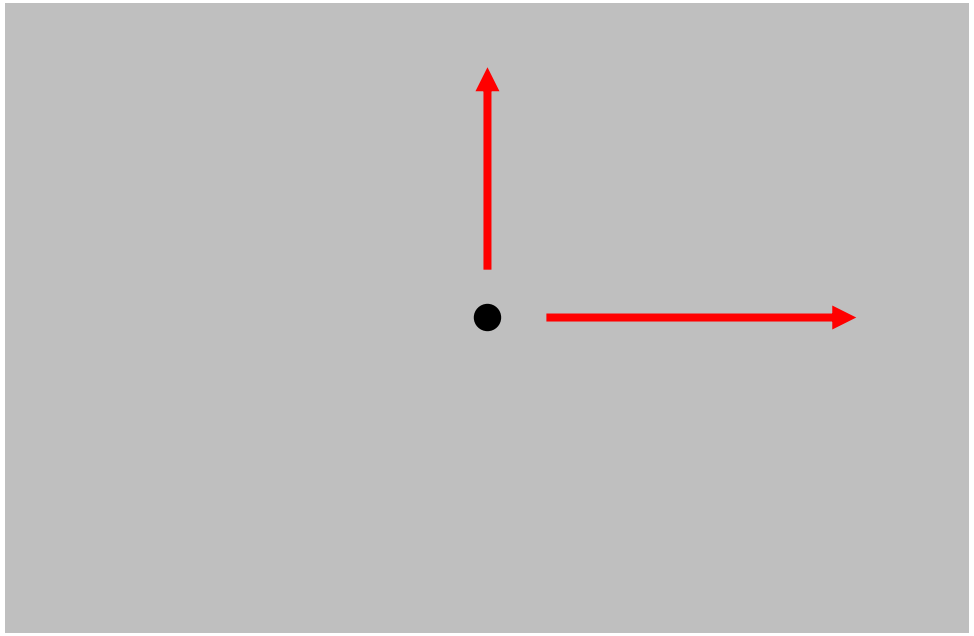
- How many degrees of freedom does this point have?



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Example: A point in 2D space

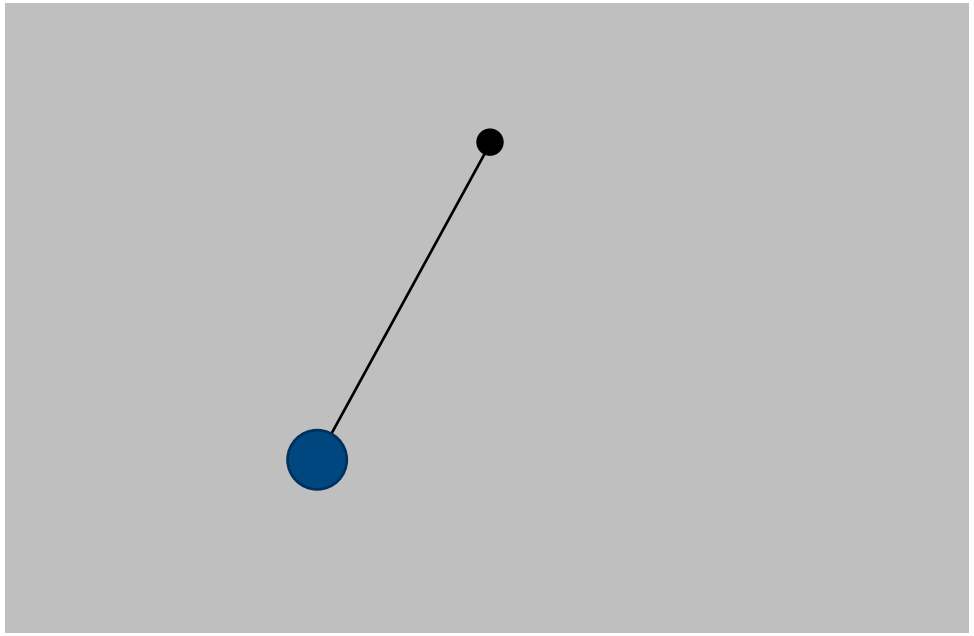
2 degrees of freedom – translation along the axes



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Example: A pendulum in 2D space

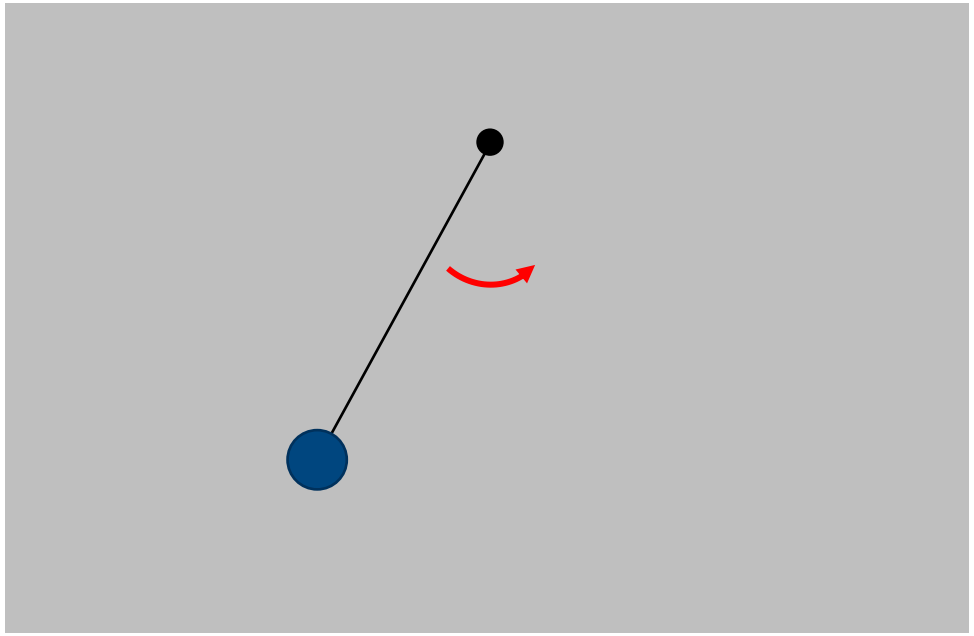
How many degrees of freedom does this system have?



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Example: A pendulum in 2D space

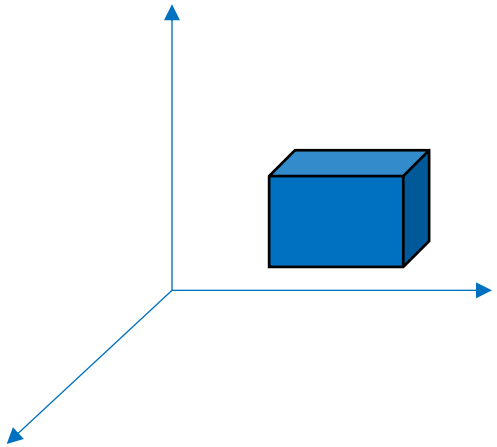
One degree of freedom – rotation about an axis through the point of suspension



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Example: A rigid body in 3D space

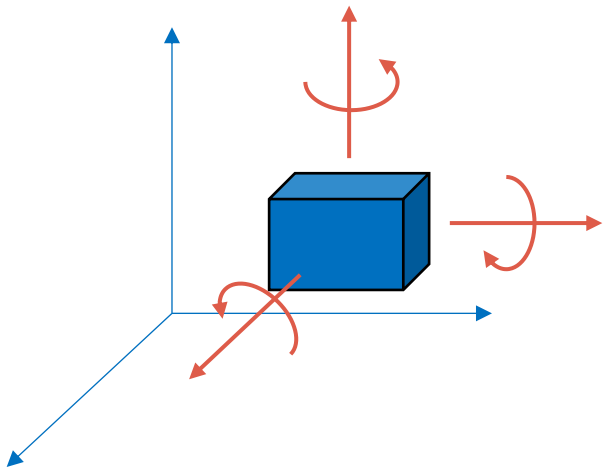
How many degrees of freedom does this box have?



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Example: A rigid body in 3D space

6 degrees of freedom – one translation and one rotation about each axis

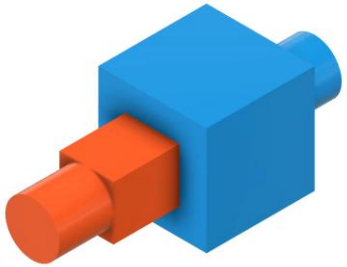


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Joints



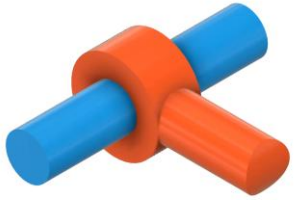
Revolute **R** : 1 DoF



Prismatic **P** : 1 DoF



Helical **H** : 1 DoF



Cylindrical **C** : 2 DoF



Universal **U** : 2 DoF



Spherical **S** : 3 DoF

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Joints constrain degrees of freedom

- Links start with
  - 6 degrees of freedom in 3D space or
  - 3 degrees of freedom in 2D space
- Joints restrict the motion of links in one or more axes by limiting translation or rotation

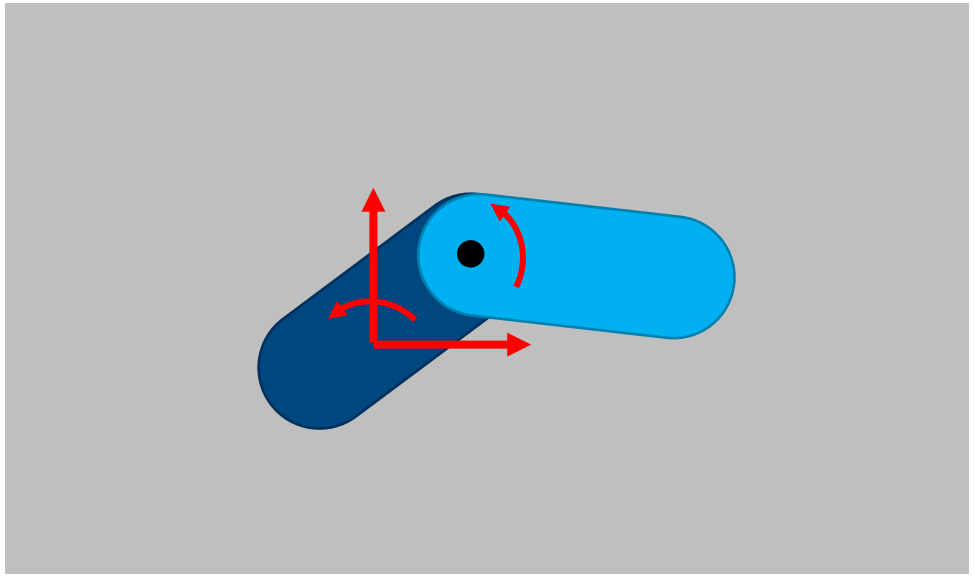
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Example: Two planar rigid bodies connected by a revolute joint

4 degrees of freedom:

The first body is free to translate in two directions and rotate about the axis normal to the plane.

The second body, if not connected can also move freely and rotate. The addition of a joint constrains its translation.



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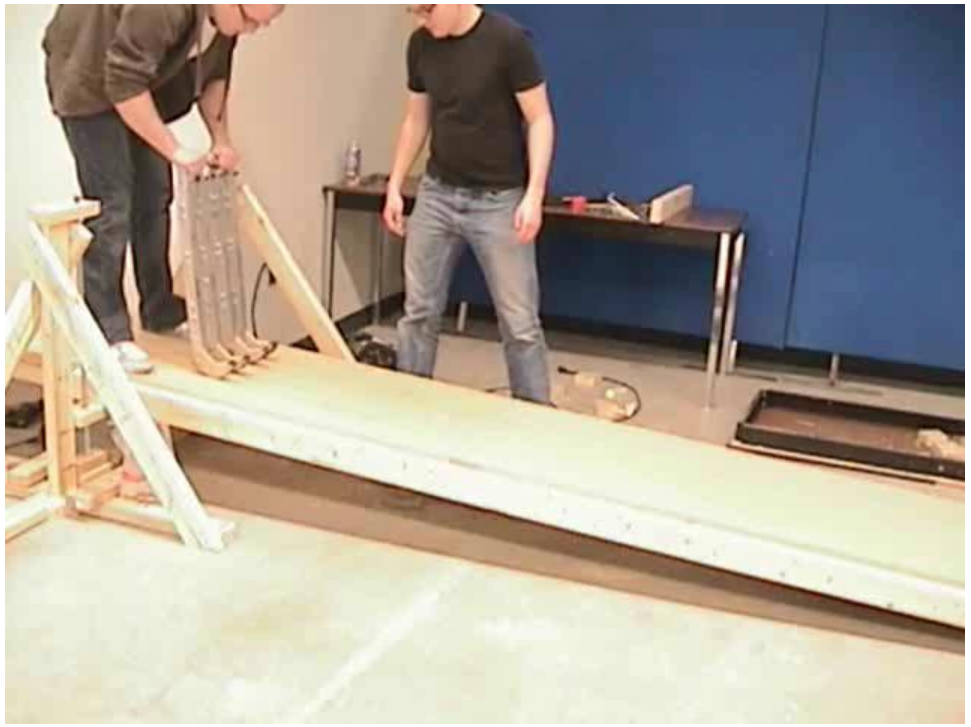
Controllable DOF vs Total DOF

- If a robot has an actuator for every degree of freedom, then all the DOF are controllable
- DOF that are not actuated are uncontrollable
- If
  - Controllable DOF = Total DOF, the robot is called fully actuated
  - Controllable DOF < Total DOF, the robot is called under actuated
  - Controllable DOF > Total DOF, the robot is called redundant

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Under-actuated systems

- Less actuators than joints.
- Low cost, simpler design, lower weight



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Mobility

Same as the degrees of freedom but is a more common term in the case of mechanisms.

"What is the minimum number of actuators that you need in a mechanism?"

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Mobility Equation: Grübler-Kutzbach equation

In 3D, $M = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - P_5$

In 2D, $M = 3(N-1) - 2P_1 - P_2$

M is the mobility / DoF of the mechanism,

N is the number of links in the mechanism (one of which is grounded), and

P_i is the number of pairs having i degrees of freedom

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Grubler-Kutzbach equation – alternative form

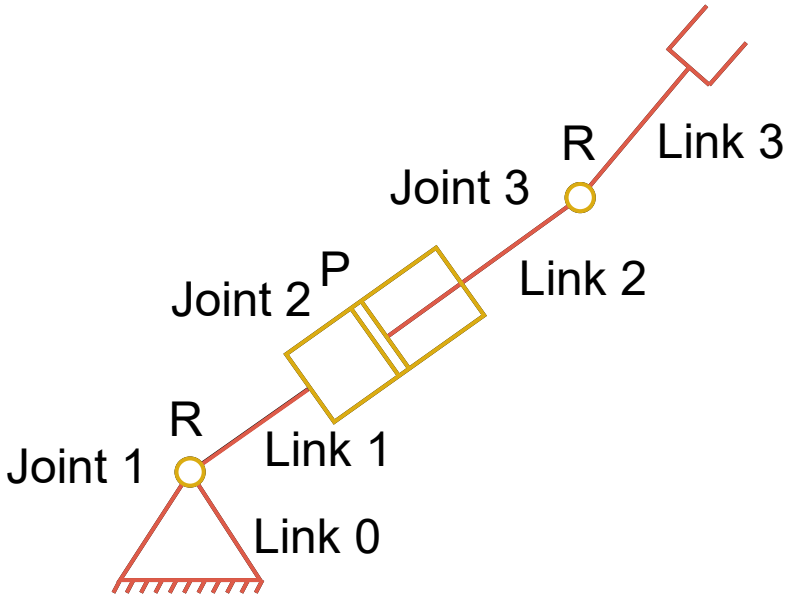
$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i \qquad = D(N - 1) - DJ + \sum_{i=1}^{J} F_i$$

M : the mobility / DoF of the mechanism,
D : 6 for 3D and 3 for 2D
N : the number of links in the mechanism (one of which is grounded)
J : the number of joints
F_i : the degrees of freedom of joint i

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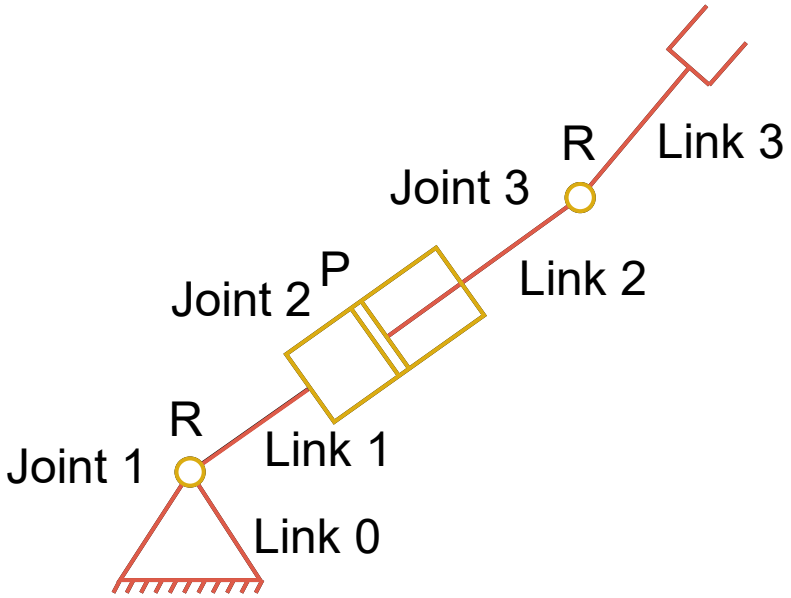
Example: Planar serial manipulator

What is the mobility of this mechanism?



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Example: Planar serial manipulator



Method 1:

$N = 4$

$P_1 = 3$, $P_2 = 0$

$M = 3(N - 1) - 2P_1 - P_2$

$= 3(4 - 1) - 2 \times 3 - 1 \times 0$

$= 3$

Method 2:

$D = 3$ for 2D

$N = 4$

$J = 3$

$F_1 = 1$, $F_2 = 1$, $F_3 = 1$

$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$

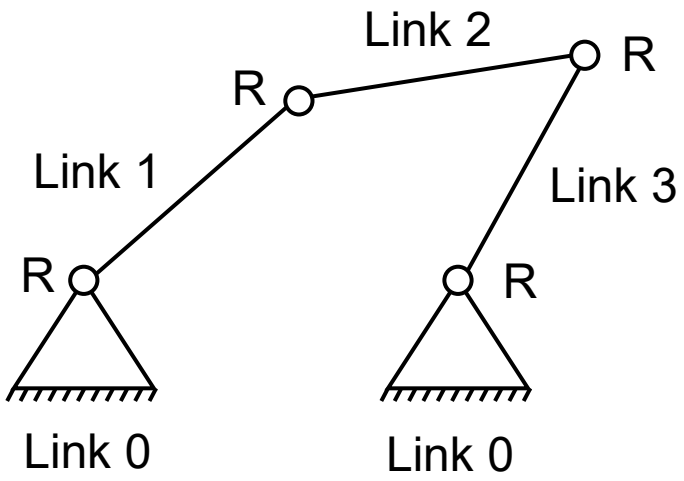
$= 3(4 - 3 - 1) + (1 + 1 + 1)$

$= 3$

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Example: Planar closed kinematic chain

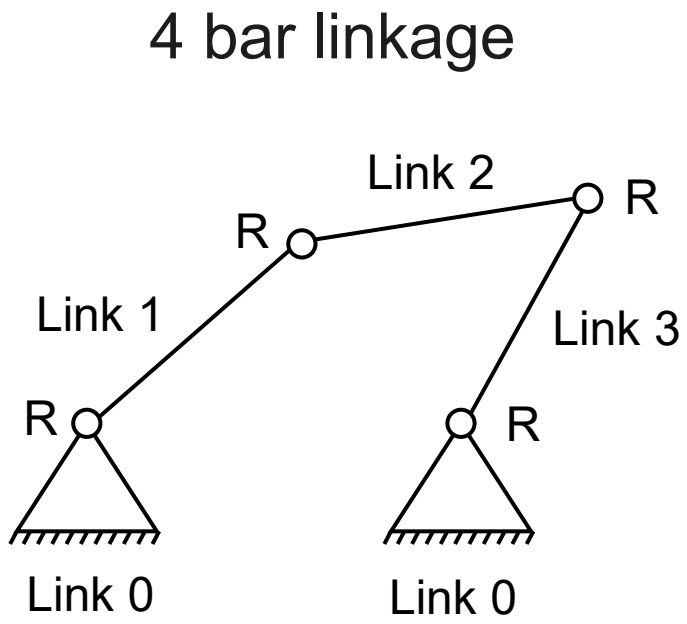
4 bar linkage



What is the mobility of this mechanism?

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Example: Planar closed kinematic chain



Method 1:

$N = 4$

$P_1 = 4$, $P_2 = 0$

$M = 3(N - 1) - 2P_1 - P_2$

$= 3(4 - 1) - 2 \times 4 - 1 \times 0$

$= 1$

Method 2:

$D = 3$ for 2D

$N = 4$

$J = 4$

$F_1 = 1$, $F_2 = 1$, $F_3 = 1$, $F_4 = 1$

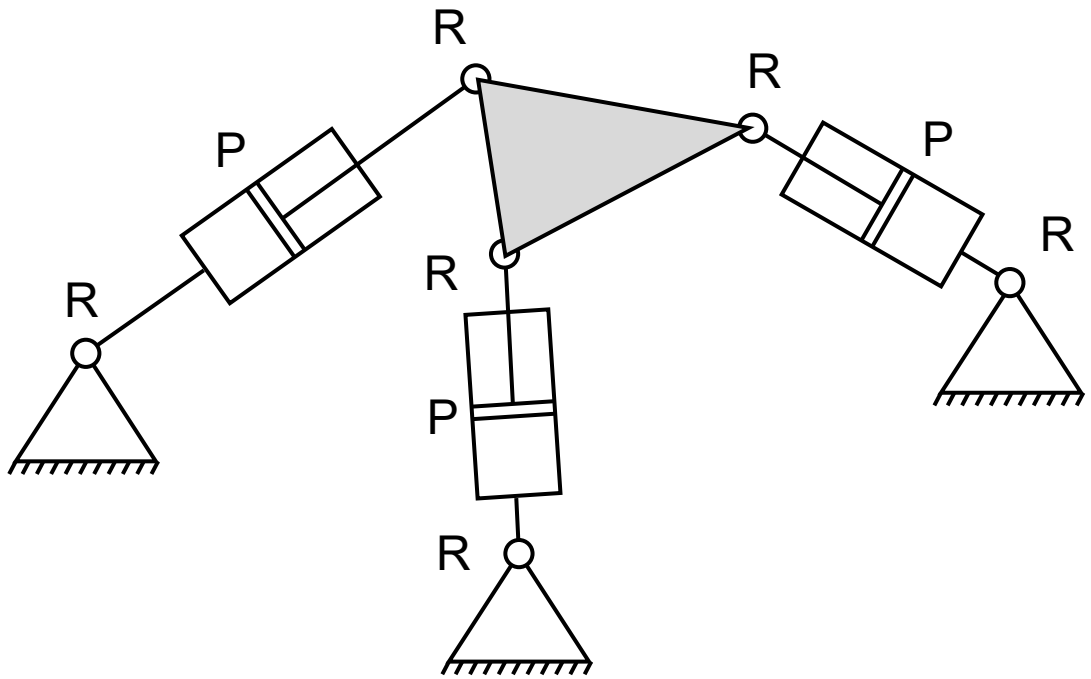
$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$

$= 3(4 - 4 - 1) + (1 + 1 + 1 + 1)$

$= 1$

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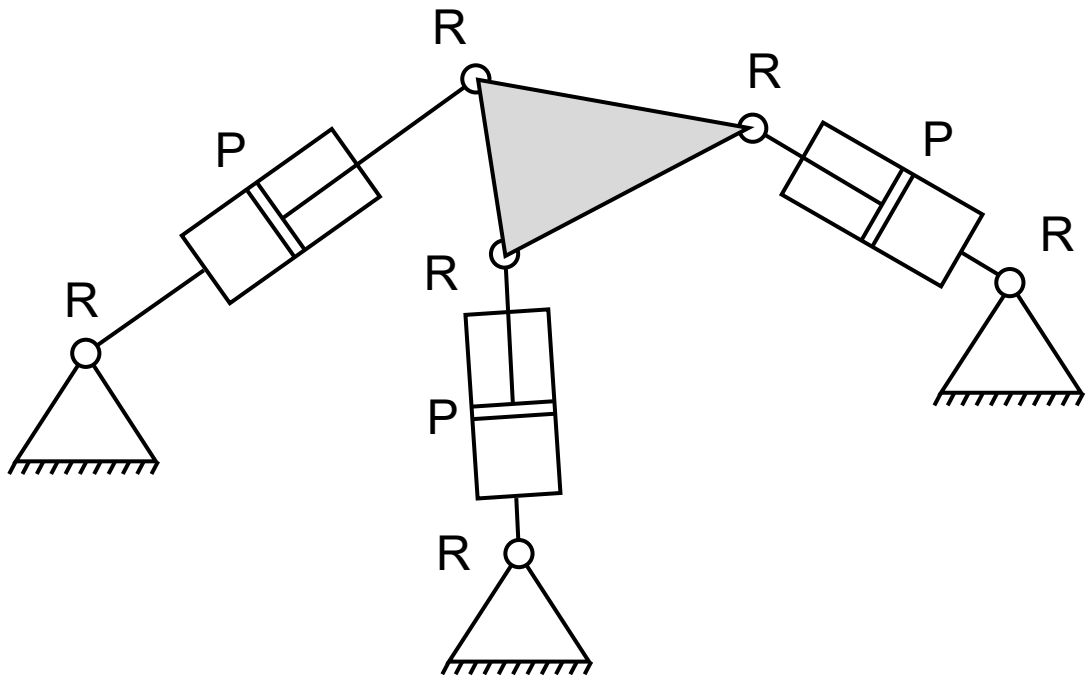
Example: Planar parallel manipulator



What is the mobility of this mechanism?

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Example: Planar parallel manipulator

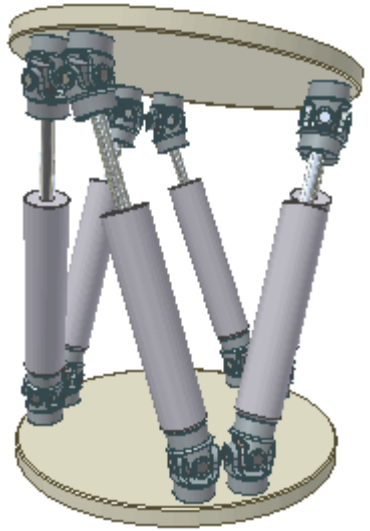


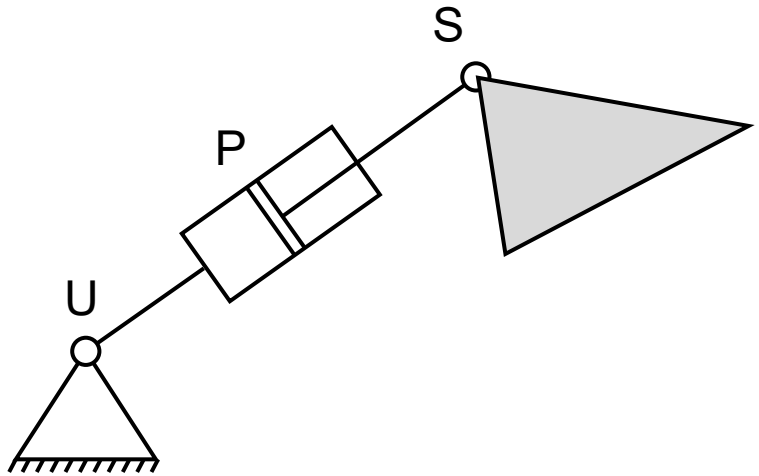
$$M = 3(N - 1) - 2P_1 - P_2$$
$$= 3(8 - 1) - 2 \times 9 - 1 \times 0$$
$$= 3$$

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$
$$= 3(8 - 9 - 1) + (9)$$
$$= 3$$

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Stewart platform – homework problem

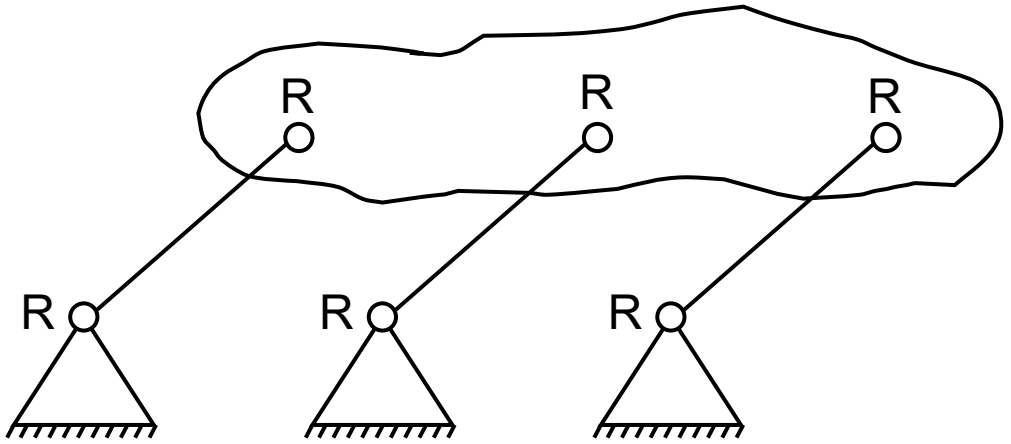




What is the mobility of this system?

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What about this system?



$N = 5$

$P_1 = 6$, $P_2 = 0$

$M = 3(N - 1) - 2P_1 - P_2$

$= 3(5 - 1) - 2 \times 6 - 1 \times 0$

$= 0$

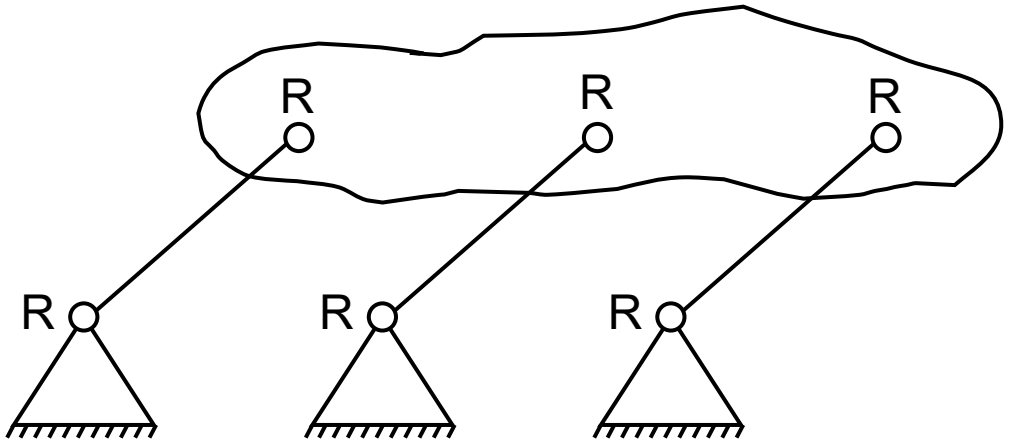
But the system can move: it has 1 degree of freedom!

The reason that the Grübler-Kutzbach equation fails is because it assumes that the constraints on motion are independent.

In this example, the function of the mechanism is not affected even if one of the middle links is removed. It is redundant and should not be counted when using the mobility formula.

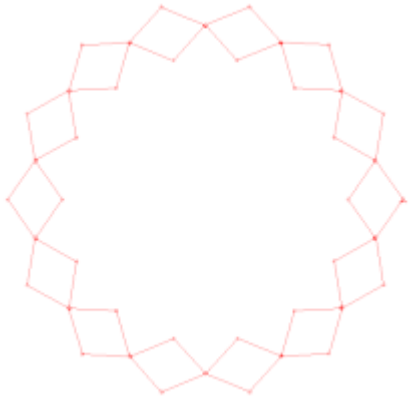
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What about this system?



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What about this one? Hoberman Mechanism



24 links and 36 revolute joints



By Pasimi - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=69639178

By Ryan Somma from Occoquan, USA - Hoberman Sphere, Chuck Hoberman 1995, CC BY-SA 2.0, https://commons.wikimedia.org/w/index.php?curid=38278791

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Workspace

The set of points that the end-effector of a manipulator can reach is its workspace.

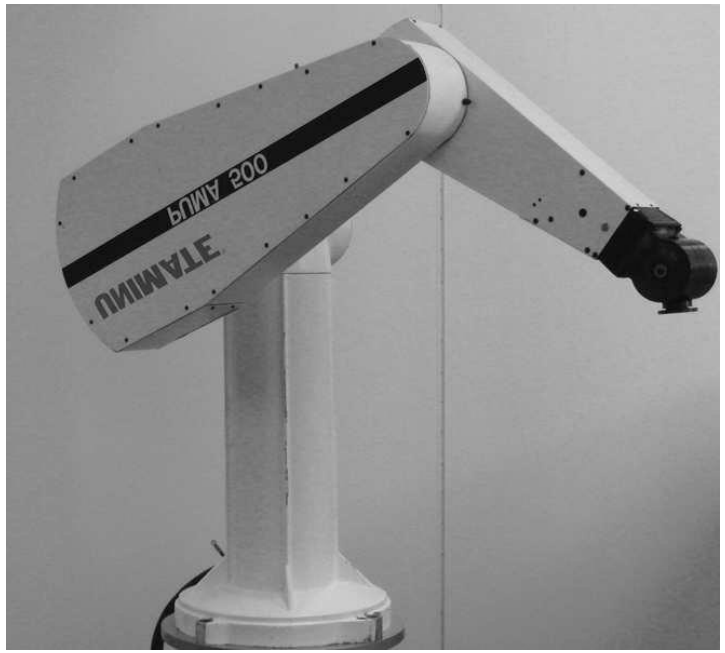
What factors determine the volume of space that the end effector can reach?

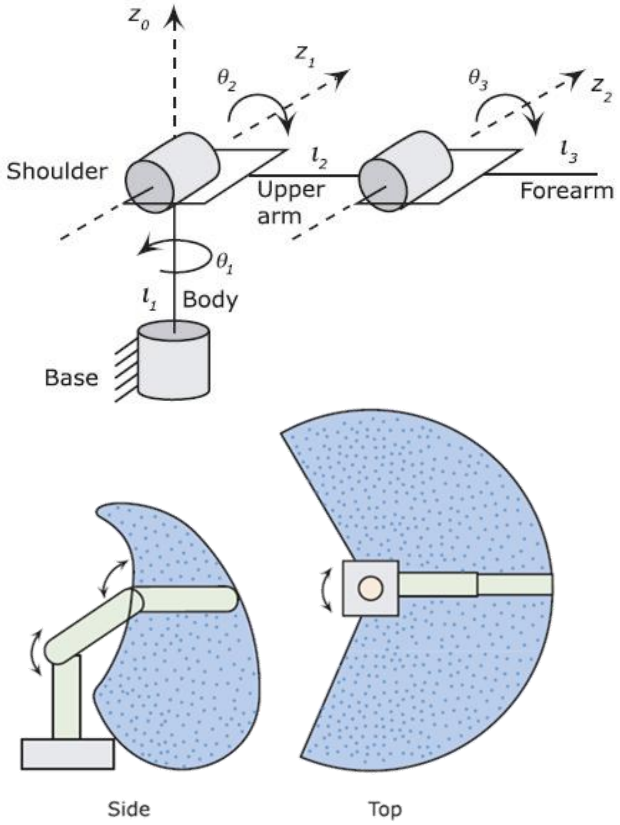
1. Geometry of the manipulator (such as the lengths of links)
2. Constraints on the joints (such as limits on the angle of a revolute joint)

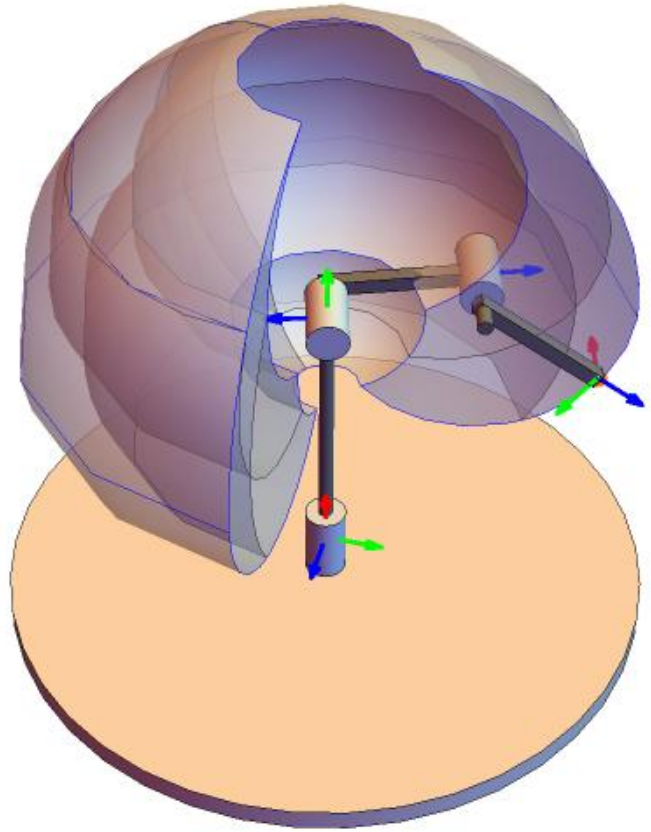
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RRR Manipulator

Programmable Universal Machine for Assembly (PUMA)





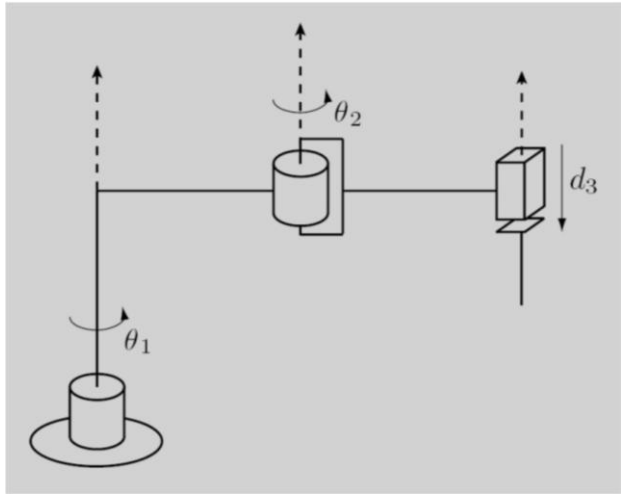


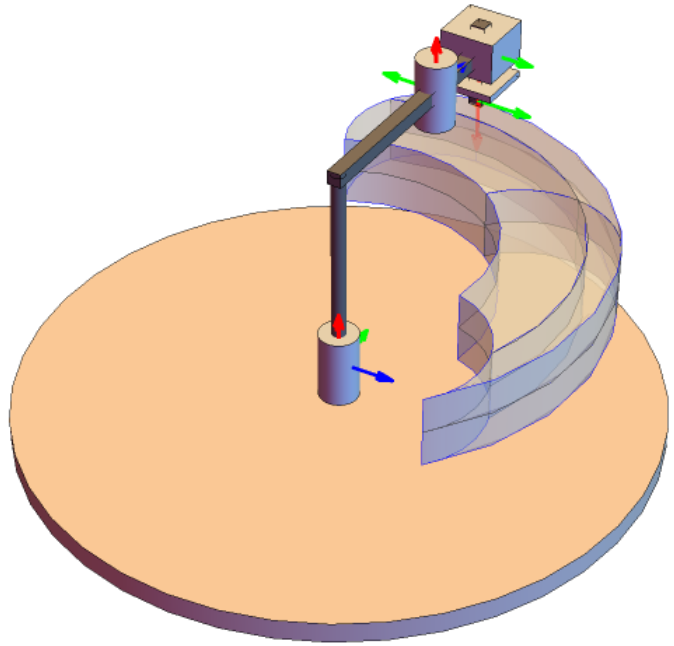
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RRP Manipulator

Selective Compliant Articulated Robot for Assembly (SCARA)

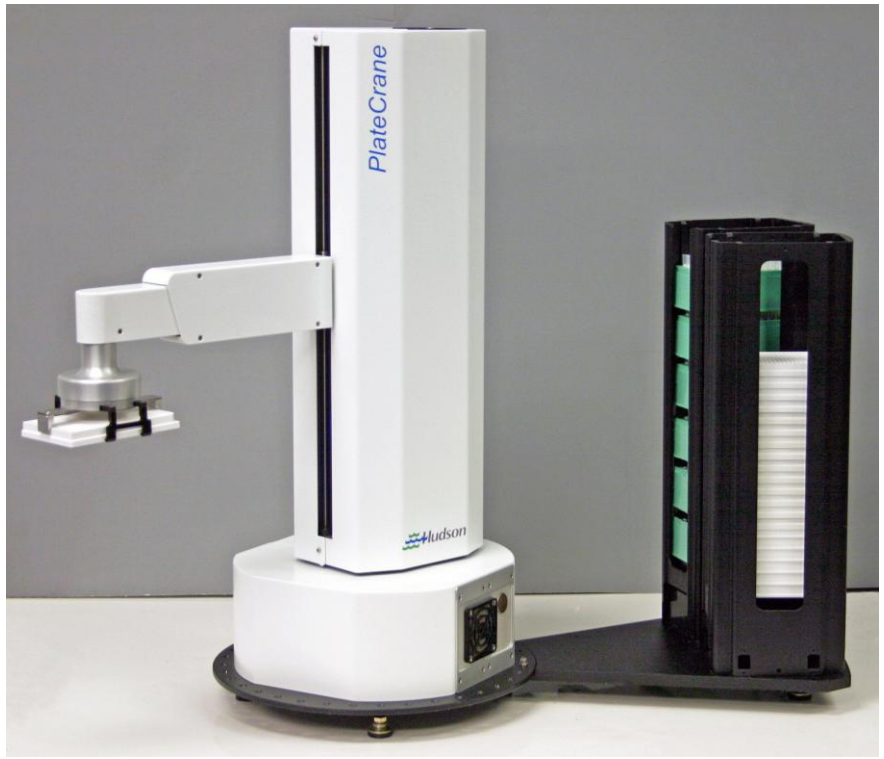




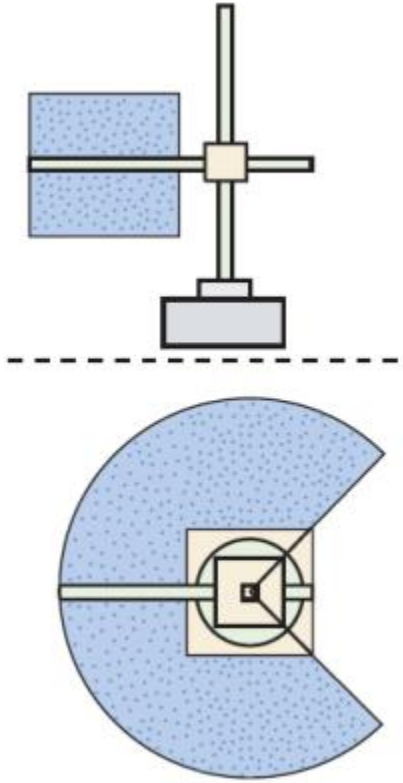


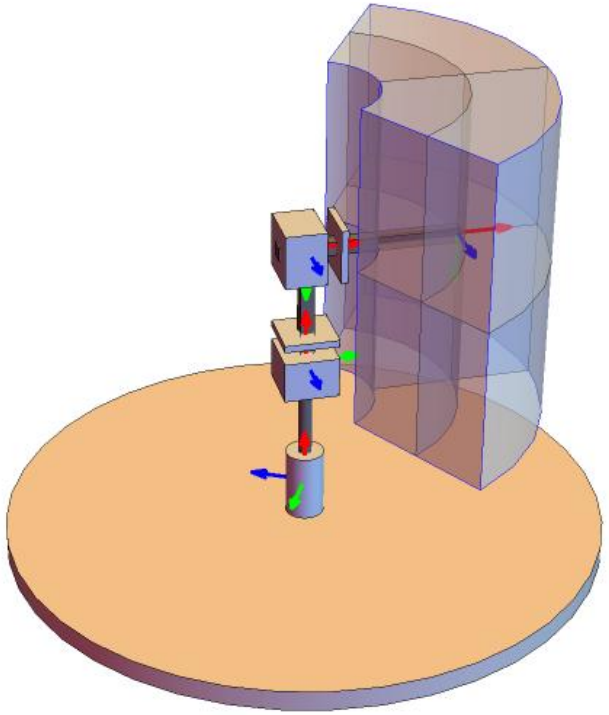
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Cylindrical Manipulator (RPP)



PlateCrane EX



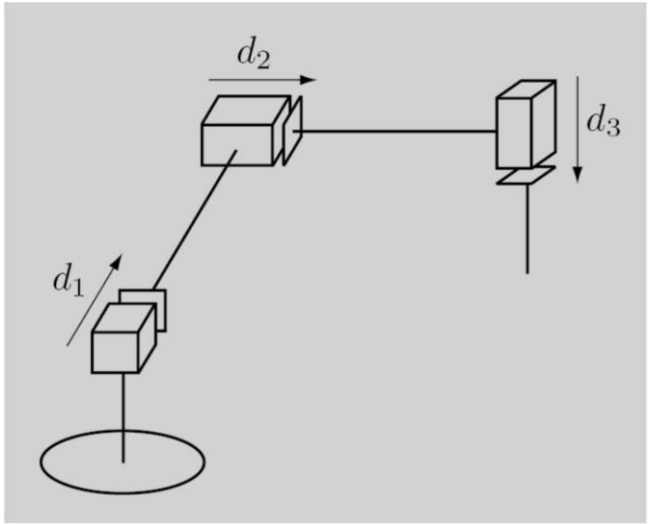


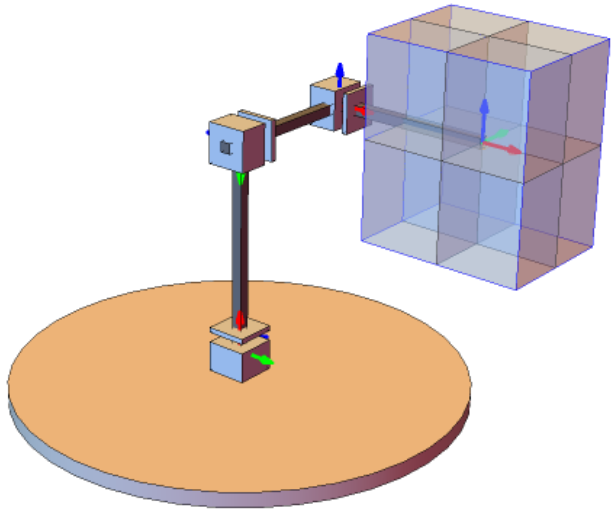
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Cartesian manipulator (PPP)



Yamaha





Reachable vs Dexterous Workspace

Reachable: the region that end-effector can reach in one orientation

Dexterous: the region that the end-effector can reach in all its possible orientations

$$W_{Dexterous} \subseteq W_{Reachable}$$

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With robot arms, we are interested in the problem of Manipulation

The goal of a manipulator robot arm is to get the end effector to a desired 3D position and orientation.

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Basic problems in manipulation

Forward Kinematics

Given the coordinates of all the joints, where is the end effector?

Inverse Kinematics

Given the tip pose – what are the joint coordinates to get there?

Velocity Kinematics

How fast is the robot end effector moving (FK) or should it go (IK) ?

Dynamics

How much torque should the motors apply to accelerate the tip?
What if the tip must apply a specific force?

Trajectory generation

Along what path should a manipulator move?
Should this be described in terms of joint positions or the pose of the tool?

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Pose

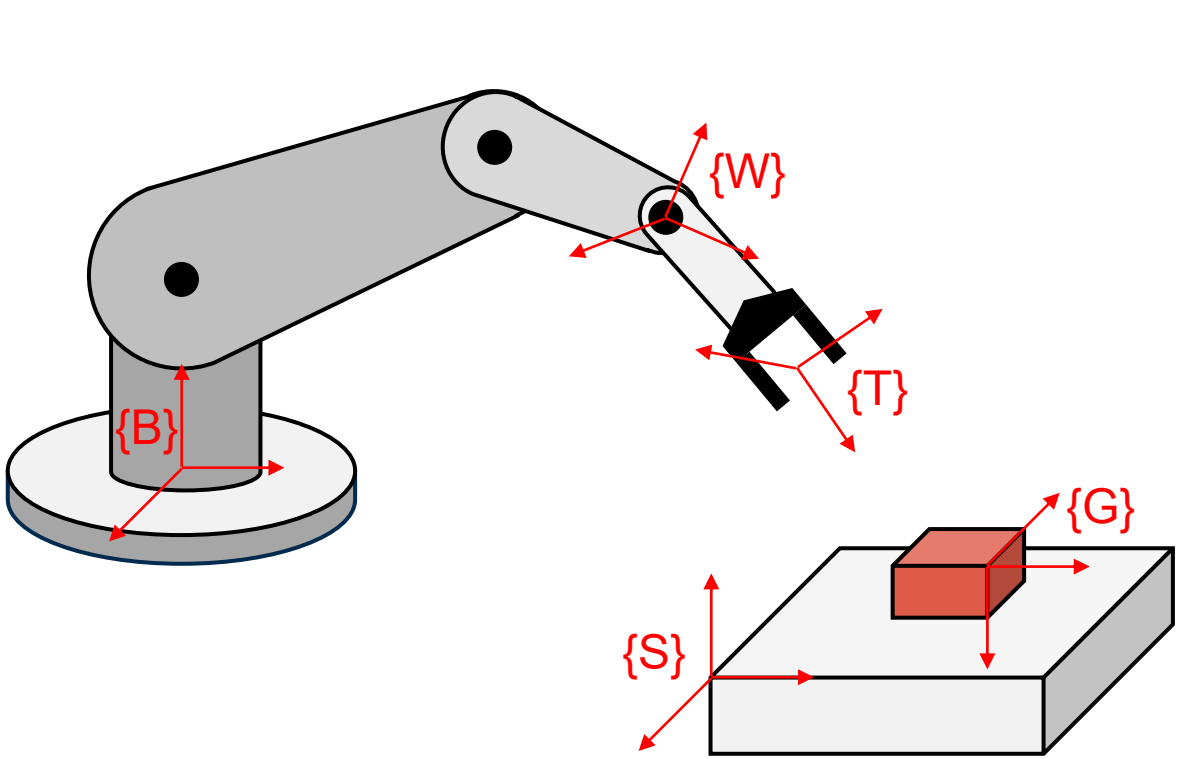
- Pose is a combination of position and orientation

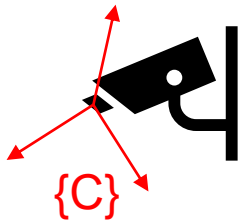
- E.g., consider a robot trying to pick a garment up from a table
  - The robot must position its gripper on the table
  - The robot must orient its gripper in a manner that facilitates pick up

- Representation of pose is important in these kinds of scenarios

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It is common to present objectives from multiple frames of reference





{B} : Base frame
{W} : Wrist frame
{T} : Tool frame
{S} : Station frame
{G} : Goal frame
{C} : Camera frame

It is helpful to know the relation between frames

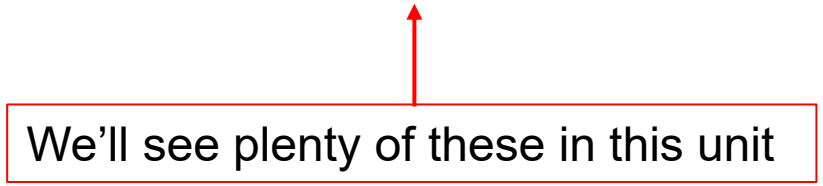
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Kinematic relation between frames

Translation and rotation relate two coordinate frames.

How many parameters do we need to relate two frames? 6 (3 translation and 3 rotation)

Mathematically, this relation can be represented as a 4x4 homogeneous transformation matrix



Recall from the last semester that there are many ways to represent relative pose.

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Summary

- Degrees of freedom define a robot, its mobility and its workspace
- Controllable vs uncontrollable DoF
- Serial and parallel robots in 2D and 3D
- Kinematics define the mapping between a robot's actuators and its end effector's position (and vice versa)
- There is a kinematic relation between coordinate frames – translation and rotation

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Next week we will study

- Coordinate transformations – revise what you learnt in EGH432
  - Homogeneous transformation matrices
  - Compound transforms
- Forward kinematics of serial link manipulators
  - Elementary Transform Sequence
  - Denavit-Hartenberg notation

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That’s all for today!

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